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Improving the Blockies User Study Web App

Shared structure, study creation and reliable data collection

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Introduction

Background

- Blockies is a framework for studying how people use AI advice and explanations
- It has three parts: a dataset generator, AI and XAI models, and a user study web app
- A user study with this framework compared low and raised stakes in a diagnosis task
- The web app is the participant-facing component that collects all study data

Goals and requirements

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- R1 Maintainability: shared code for all studies
- R2 Reusable functions: common study functions
- R3 Study creation: create a study from a CSV file and images
- R4 Data accuracy: store every response correctly
- R5 Testing: automated tests for the database
- R6 Components: reusable interface parts
- R7 Configurable layouts: layouts set by configuration

Starting situation

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- Every study had its own copy of the HTML, CSS and JavaScript files
- A new study required copying files and editing JSON by hand
- Layouts were fixed in the code
- No tests checked that results reached the database
- The Docker build failed because of a GLIBC version mismatch

Implementation

Before and after

Before

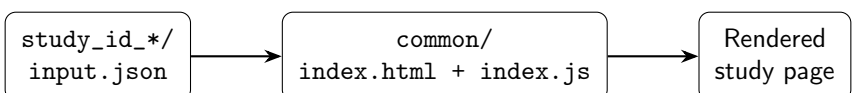
- Each study folder had its own `index.html`, `index.css` and `index.js`
- Every change had to be repeated in all studies

After

- One `common` folder with the shared files
- Each study keeps only `input.json` and its images

One shared client for all studies

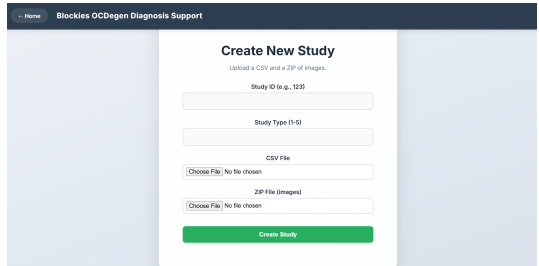
- The duplicated files were replaced by one client in `src/client/common`
- The file `input.json` describes the layout and the data of the study
- The client reads it and builds the page, so updates are made in one place



Creating a study in the browser

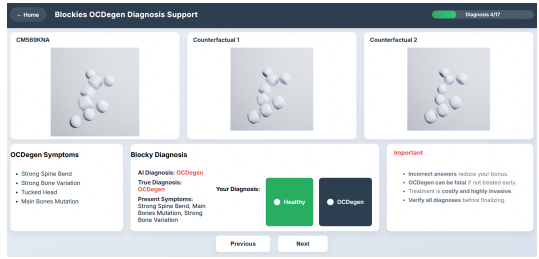
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- The user uploads a CSV file and a ZIP file of images on a new page
- The server reads the CSV and extracts the images
- The server creates `input.json` with a suitable layout
- The study is added to the database and is ready to use



Connection to the dataset generator

- The Blockies generator produces images and a CSV file
- The server detects the columns `Shap`, `Concept1_Salience` and `Example1`
- From these columns it builds the correct layout automatically
- Study types 2 to 5 were created this way, without editing `input.json`

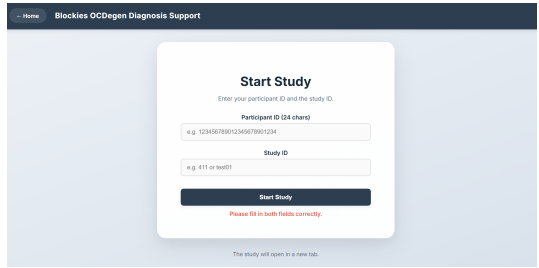


Data integrity and testing

- Incoming results are checked before they are stored
- A unique index prevents duplicate answers
- Timestamps are updated automatically
- Automated tests (Mocha, Chai, Supertest) cover the main endpoints
- All Mocha tests pass

Interface

- The study page is responsive and adapts to different screen sizes
- A progress bar shows how far the participant has come
- Keyboard navigation: arrow keys, Enter and Escape
- The diagnosis card is revealed by a click



Docker deployment

- The Docker build failed because of a GLIBC version mismatch
- The cause was the Node base images based on Alpine and Debian slim
- Changing the base image to `ubuntu:24.04` fixed the problem
- Node.js 18 and the build tools are installed in the image
- The app now starts in a container

Results

Results against the requirements

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- ✓ R1 Maintainability: one shared client for all studies
- ✓ R2 Reusable functions: common helper functions
- ✓ R3 Study creation: studies are created from a CSV file and a ZIP file
- ✓ R4 Data accuracy: validation, unique index and timestamps
- ✓ R5 Testing: automated tests for the main endpoints, all passing
- ✓ R6 Components: page built from reusable cards
- ✓ R7 Configurable layouts: layout set in `input.json`, one to five images

Live demonstration

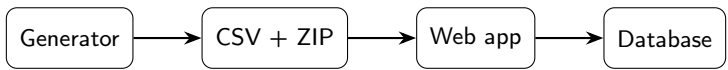
- The web app runs locally at `http://localhost:7000/`
- Home page
- Create a study from a CSV file and a ZIP file
- Start a study and navigate through the pages
- Submit a diagnosis and check the stored data

Discussion & Conclusion

Discussion & Conclusion

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- The web app is now easier to maintain and extend
- Integration with the Blockies dataset generator is a key achievement
- A study can be created from generator output in minutes
- Next: connect the AI and XAI model outputs more directly
- Next: consider a component framework and add interface tests



Thank you

Questions and discussion